



# ForgeGIS Studio<sup>TM</sup>

*An Operator Surface That Stays on One Host*

ForgeGIS Studio<sup>TM</sup> is the browser workspace for the Forge suite: a live map, a pipeline canvas, a batch runner, and a natural-language agent acting on the operator's own map. Its deployment model is a single operator on a single machine, and that model is defended by default — the server binds to loopback, every path a caller supplies is sandboxed, resources are capped, and the agent reaches the map through its own authenticated door.

## 127.0.0.1

### loopback by default

*rebind it outward and Studio warns on every boot while it is off loopback*

## Sandboxed Separate

### every caller-supplied path

*uploads confined and extension-checked; opted-in roots canonicalised, deny-listed and session-scoped*

### credential for the agent

*the agent acts on the map through a token-validated bridge, not the browser's cookie*

### Why single-host is a decision

Studio's browser API carries no login because it is not meant to be reached by anyone but the operator at the machine. The prerequisites for any hosted deployment — real authentication, CSRF protection, rate limiting, TLS, an audit log and a sandbox re-review — are named in its design, not implied, so scaling outward is a planned step rather than a shortcut.

### Bounded by construction

Upload and pixel caps, a GPU-memory estimate checked before any operation is dispatched, a bounded executor pool that refuses rather than thrashes, and a per-operation timeout cap the worst case one workstation has to absorb. Sessions expire, cross-site writes are refused, and zip and KMZ uploads are vetted before extraction and capped on real bytes as they inflate.

### An agent you can bound

ForgeMind acts on Studio's map only through the authenticated `studio-mcp` bridge, under an operator-only sandbox that cannot reach paths a browser session added. On ForgeMind's compiled path a request becomes a validated pipeline or a named refusal, and every named result the agent places is recorded in the catalog.

## WHAT PROGRAMS GET

### ***Terrain and analysis workflows, composed and batched***

Hillshade, slope, viewshed, least-cost paths and the rest of ForgeGIS's raster and vector operations become nodes on a canvas; a graph proven on one area runs across a folder of inputs in batch, with each tile's GPU memory checked as it is dispatched and a durable report at the end. Pipelines converted from QGIS models carry their import warnings, including the one case that runs but answers differently.

### ***The operator's own data***

Load from the ForgeData catalog — including a shared, curated catalog for the whole workstation — or drop KML, KMZ and zipped shapefiles straight onto the map. Area-windowed reads pull just the pixels you asked for, and a registered-but-missing dataset is caught when you pick it, not when it fails to read.

### ***Large results, stated honestly***

When a Tools-panel geometry result is too large to inline, Studio draws it from the engine's file if it is geographic and shows a named notice if it is not — it never draws projected coordinates as if they were degrees. The agent's answers are drawn on the operator's map, alongside the text of its reply.

**Footprint.** An opt-in component of the Forge suite, installed and started by the Seaglass Control Center™ on one Windows x64 workstation, on port 8081 behind loopback. The suite brings its own Java runtime; the agent bridge is a standalone executable. The basemap and URL layers need a network; rasters render offline.

## **Request the Technical Brief and an evaluation build**

The Technical Brief covers the architecture, the security posture in full, the batch runner, the agent bridge and deployment. ForgeGIS Studio is part of the Forge suite and is not sold separately. The Forge suite is commercially licensed; evaluation builds are available to qualified programs.

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